

[813 CHALLENGE // URBAN EXPANSION, LAND USE CHANGE & HEAT RISK]

CoolCairo

Which blocks should Cairo cool first?

Summer heat risk mapped block by block from satellites, and a 3D tool where planners paint cool roofs, trees, cool pavements and parks and see the effect live.

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EGYPT

Cairo's surfaces pass 45 °C every summer

45.7 °C

a typical east-Cairo block's summer surface temperature (Landsat, 2023–25)

95,141

residents in our 2.2 × 2.2 km Nasr City district (WorldPop 2024)

+1 °C

new blocks built 2016–23 run hotter than established ones (46.9 vs 46.0 °C)

Who is affected: residents of dense blocks on hot surfaces, old and new. Heat hits older people, young children and outdoor workers hardest.

We want to **map** summer heat risk (surface temperature, the materials driving it, and the residents exposed) in **fast-growing MENA cities**, starting with **Nasr City, Cairo**, so that **city planners** can decide which blocks to prioritise for cool roofs, street trees, cool pavements and pocket parks.

End user: city planners cooling the blocks where people live

01 SEE

Where is it hot?

Surface heat and surface materials per 90 m block, in 3D.

02 RANK

Who is exposed?

Heat risk: residents × °C above a typical block.

03 PAINT

Try a measure

Cool roofs, street trees, cool pavements, pocket parks, painted onto blocks.

04 COMPARE

What changes?

Live before → after: temperature, heat exposure, residents cooled.

WHO ADOPTS IT

City and governorate planning departments, new-city developers, and climate-adaptation programmes that fund cooling.

WHY IT MATTERS

Cooler surfaces where people live, for health and the environment: a third of the blocks, chosen by risk, removes 74% of heat exposure.

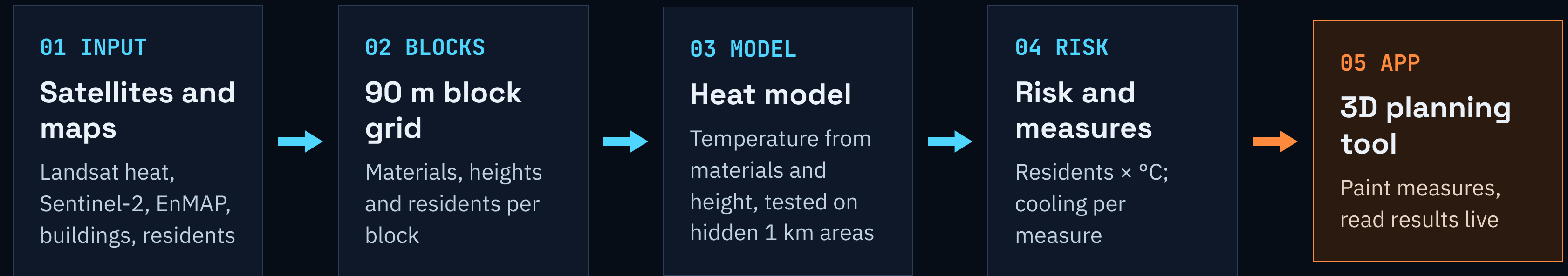
Free satellite data that covers the whole region

Source	What we took	Used for
Landsat 8/9 (USGS)	65 summer scenes, 2023–25	Surface temperature, median per block
Sentinel-2 L2A (ESA)	70 granules (58 days), 10 m	Surface materials, vegetation
EnMAP L2A (DLR), hyperspectral	1 scene, 22 Apr 2025, 30 m	Which surfaces drive heat
OpenStreetMap, Google Open Buildings 2.5D	~1,900 buildings, heights, 2016–23	3D city, roofs, urban growth
WorldPop 2024	100 m population grid	Residents per block

Streamed from Microsoft Planetary Computer and DLR EOC Geoservice. No paid data; raw EnMAP data stays out of the repository (licence).

Why EnMAP, not Satellite 813: 813 imagery of a team's own area comes in the incubation phase (organizers, 1 Oct 2026), so the PoC uses EnMAP; 813 data for Cairo is our first incubation step.

From satellite scenes to a planning tool



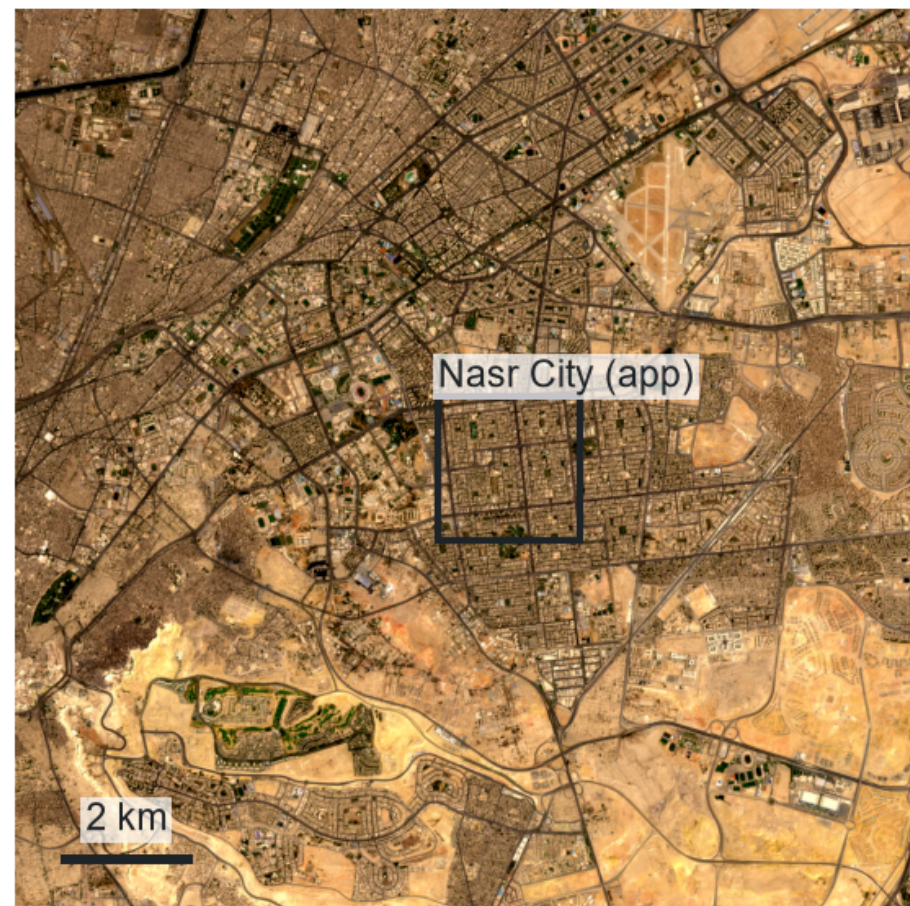
PYTHON 3.12 • uv • NOTEBOOKS 01-06 • UNIT TESTS

UNITY 6 APP

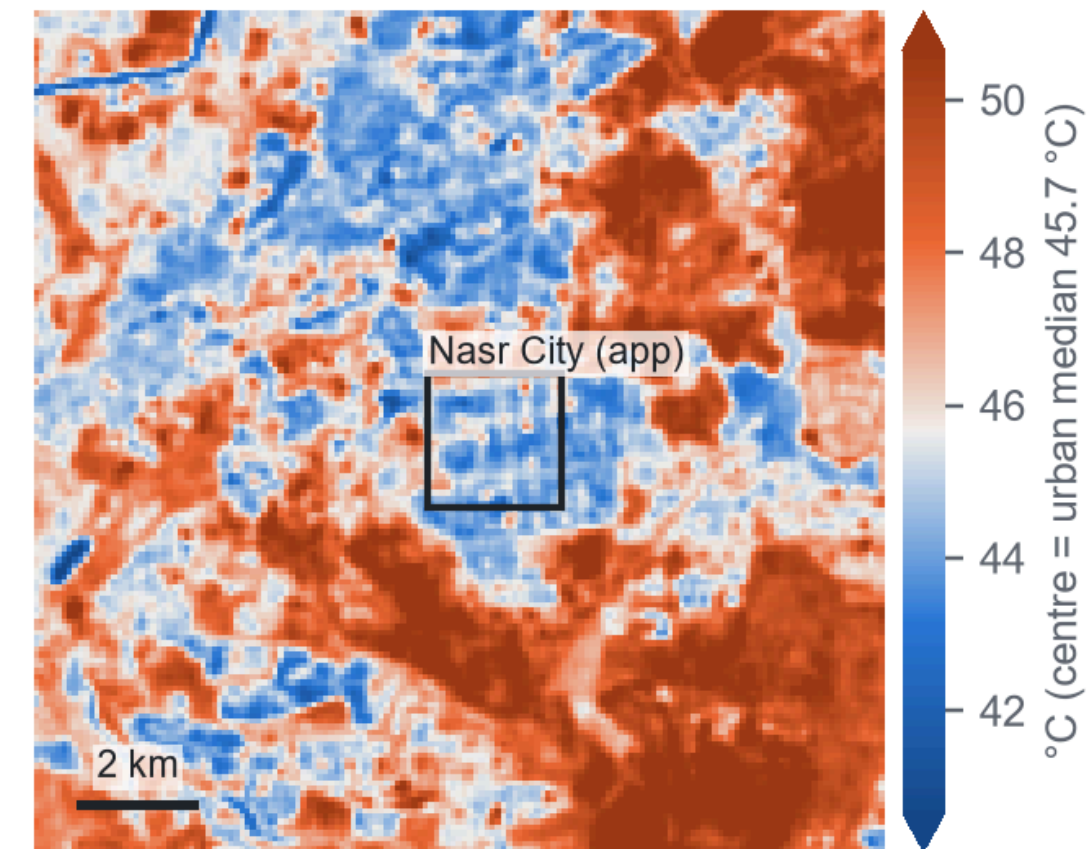
The analysis writes one file, [export/district.json](#); the app evaluates the same model live, so every number on screen traces back to a notebook.

Desert hotter than the irrigated city

a Model area, Sentinel-2 true colour



b Summer surface temperature, 90 m blocks

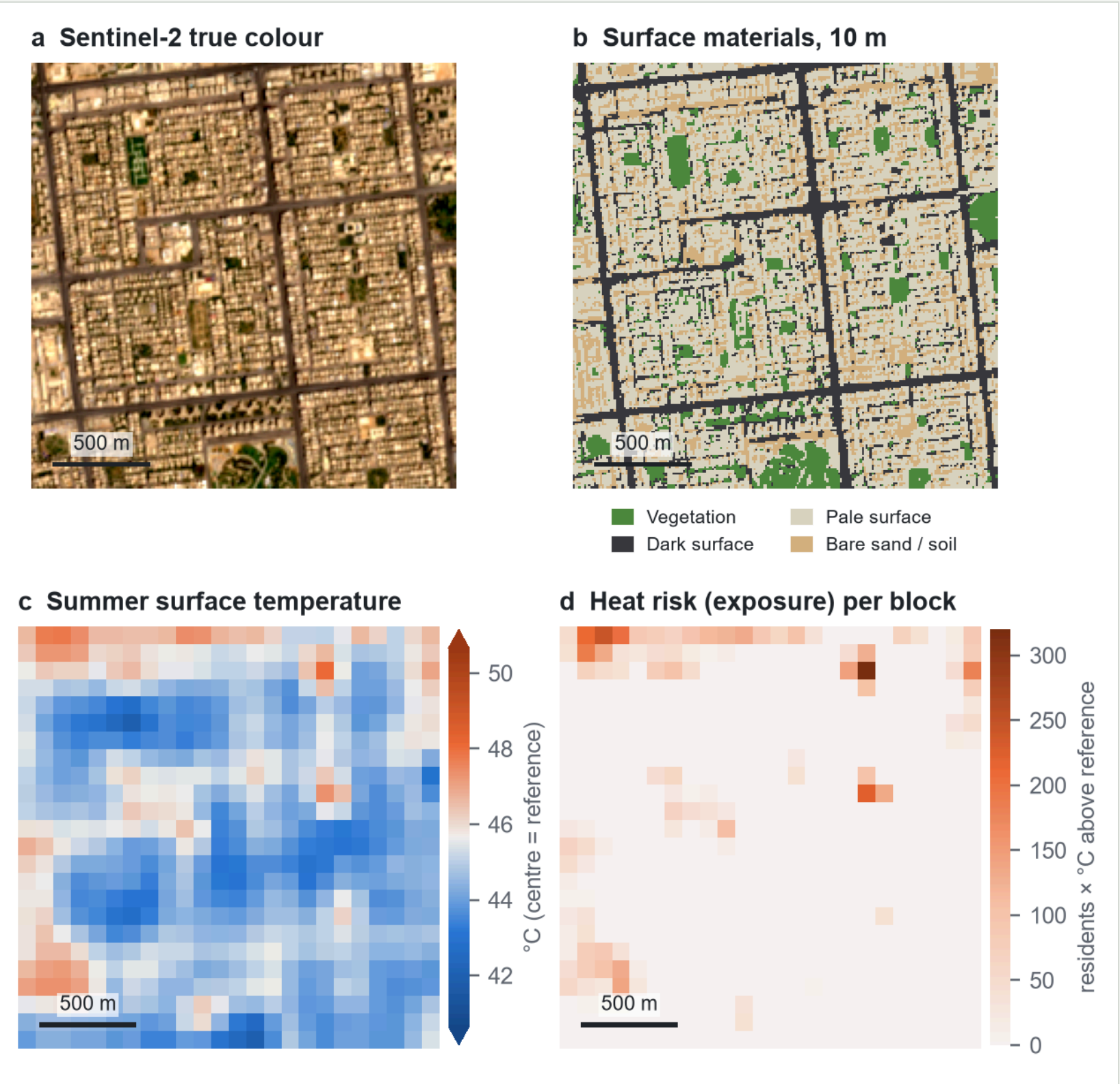


(a) Sentinel-2 summer photo; **(b)** Landsat surface temperature per 90 m block, median of 65 summer scenes, centred on the typical block (45.7 °C). Box: the Nasr City district in the app.

Heat and heat risk are not the same map

Nasr City from the analysis: (a) Sentinel-2 summer photo, (b) surface materials at 10 m, (c) summer surface temperature per 90 m block, (d) heat risk, residents × °C above the typical block.

Bare desert edges are hottest, but few people live there.
Risk concentrates where dense housing meets hot surfaces, so those blocks come first.



Hyperspectral EnMAP explains heat best

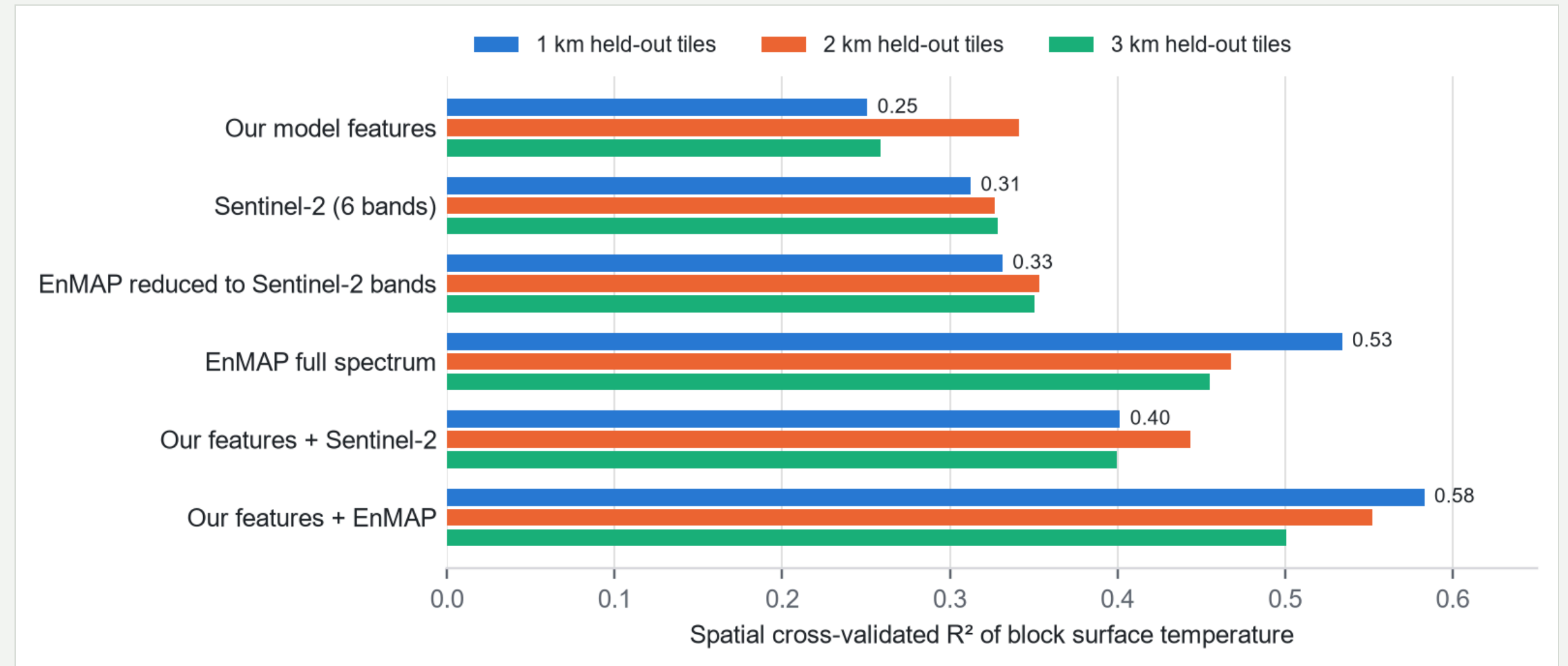
R² WITH OUR FEATURES

0.40

Sentinel-2 bands

0.58

EnMAP hyperspectral



Hidden 1 km areas; holds at 2 and 3 km. Spectra alone: EnMAP 0.53 vs 0.33 for the same scene cut to Sentinel-2's bands, so the gain is spectral detail. No gain for telling buildings from desert.



The app today: Surface heat view, summer surface temperature per 90 m block over 3D buildings. Left: views and controls; bottom: the four cooling tools; right: the colour key.



10 // OUTPUT · THE TOOL, AFTER A PLAN

44.3 °C

mean block surface, was 44.7

1,686

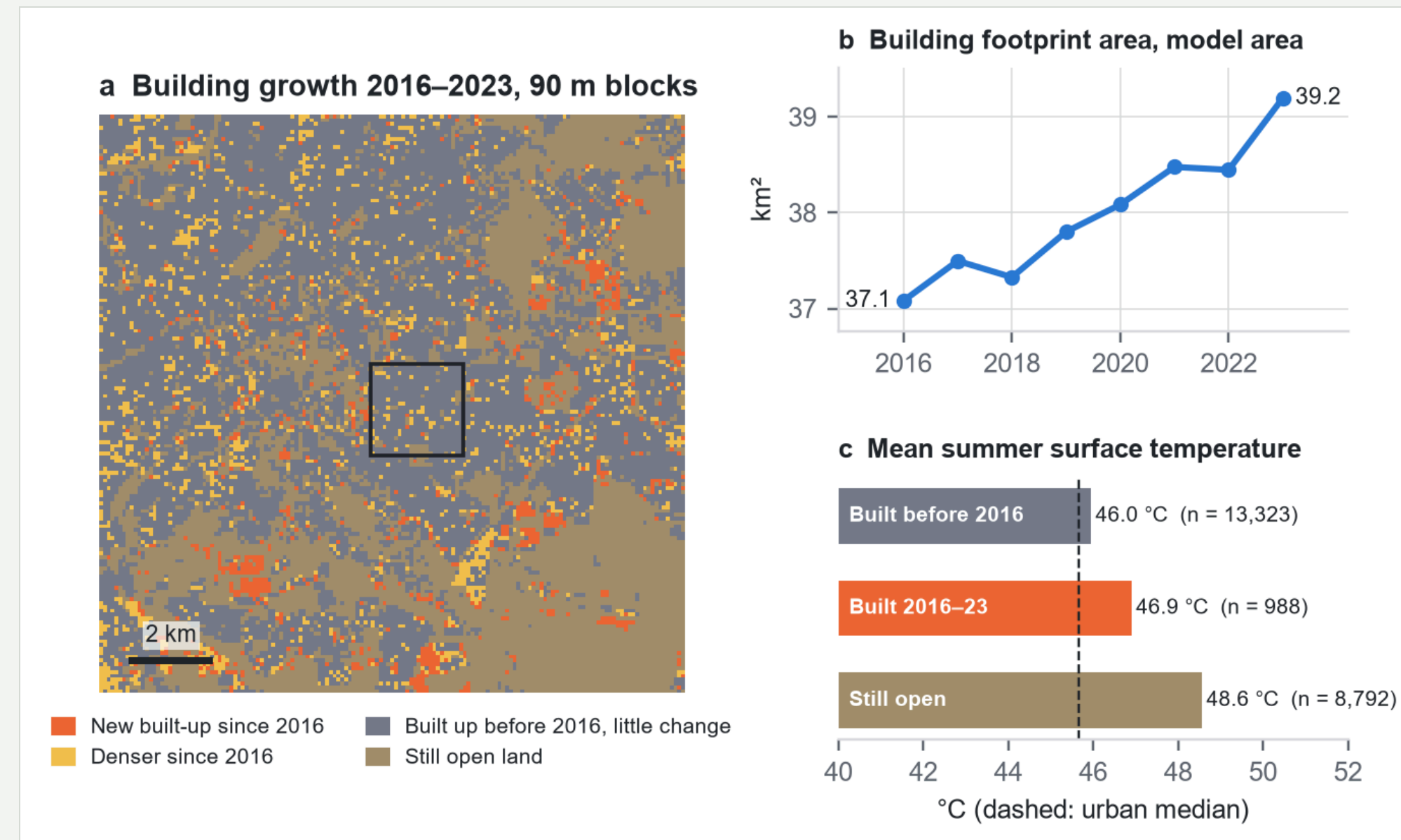
heat exposure, -74% vs today

22,588

residents in cooled blocks

After a plan: cool roofs and pocket parks painted on the riskiest third of blocks. Heat risk view: only a few blocks are still above the typical temperature. Coloured rings mark every treated block.

New neighbourhoods are being built hot



+5.7%

building footprint in east Cairo, 2016–23 (37.1 → 39.2 km²)

988

blocks turned from open land to built-up, now home to ~61,000 people

46.9 °C

new blocks, vs 46.0 established and 48.6 open desert

What we checked, and what we cannot claim

R^2 0.25

Heat model on hidden 1 km areas, ± 1.2 °C, 2,538 blocks

16 / 18

material spot checks agree with satellite imagery (89%, not blind)

r 0.60

building heights vs OSM tags, bias -1.4 m, error 6.2 m

51 / 51

automated checks pass inside the built app

1 Surface, not air. Satellites measure how hot surfaces get.

3 Model explains ~a quarter. Fine for comparing measures, not exact predictions.

5 OpenStreetMap gaps. The model learns from well-mapped blocks.

7 One EnMAP scene (April). A strong signal, not yet the main model.

2 90 m blocks only. The heat sensor cannot see single buildings.

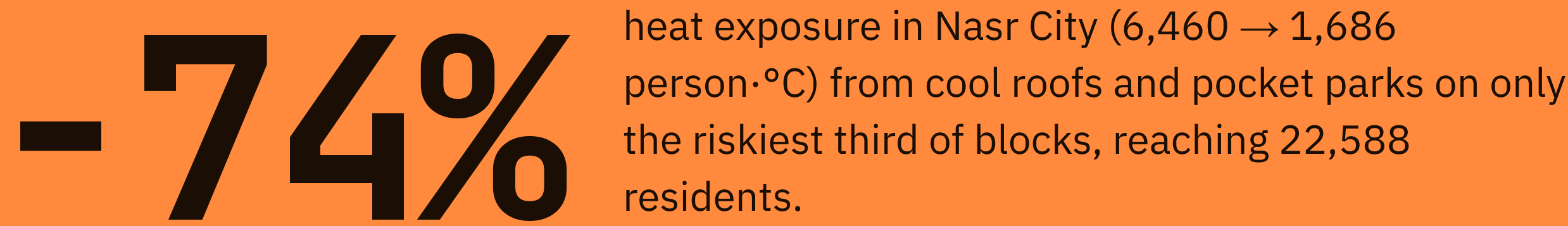
4 Dusty roofs vs sand, shadows. Why cool roofs use published physics.

6 Modelled residents. WorldPop; no age or housing vulnerability yet.

How we would overcome each limitation

Limitation	How to overcome it	Effort	Evidence it helps
1 Surface, not air temperature	Weather stations on the same days; 10–20 air sensors for one summer; SOLWEIG comfort model	1–2 days; sensors: incubation	Direct test: hot surface = hot air?
2 Block level only (90 m)	Sharpen to ~30 m with Sentinel-2; ECOSTRESS (~70 m, day and night); ESA LSTM, TRISHNA (~50 m)	Days; missions when they fly	Published; night heat matters most
3 Model explains ~25% (R^2 0.25)	Add EnMAP spectra, neighbouring blocks, desert distance, street width; test a non-linear model	Half a day	EnMAP measured: R^2 0.40 → 0.58
4 Dusty roofs vs sand; shadows	Classifier on all 6 Sentinel-2 bands; shadows from heights and sun angle; blind spot check	About 1 day	Test pixels: 76% → 86%
5 OpenStreetMap gaps in Cairo	Google Open Buildings or Microsoft footprints as the roof map; add missing buildings to OSM	1–2 days	More than today's 2,538 training blocks
6 Modelled residents; no vulnerability	WorldPop age grids (over-65s, under-5s), CAPMAS census, clinics and shade → heat vulnerability index	1–2 days; index: incubation	Standard public-health method
7 One EnMAP scene (April)	Request summer EnMAP scenes; NASA EMIT, Tanager, PRISMA; Satellite 813 data in incubation	Days to months	One scene already: 0.40 → 0.58

Targeting beats spreading: cool the right third of blocks



ONE MEASURE IN EVERY BLOCK, FOR COMPARISON



SDG FIT

SDG 11 Sustainable cities

SDG 13 Climate action

SDG 3 Good health

Heat-risk planning as a service

CUSTOMERS

City and governorate planning departments, new-city developers, and climate-adaptation funders (development banks, NGOs) that pay for cooling programmes.

PRODUCT AND DELIVERY

Today: the desktop app plus a reproducible pipeline. Next: a web version on GIQ. Each new city uses the same free data and code; deliverables are ranked block maps and costed cooling plans.

REVENUE

A setup fee per city plus a yearly licence for updates and new summers; paid pilots; plan reports for funded cooling programmes.

COSTS AND WHY IT PAYS

All input data is free (Landsat, Sentinel-2, OSM, WorldPop); costs are mainly team time and light cloud compute. For the client: in Nasr City, a third of the blocks removes 74% of heat exposure.

First step: a pilot with one Cairo district authority or new-city developer during incubation.

From a Nasr City prototype to a MENA heat-planning tool

Validate on the ground

10–20 air-temperature sensors in Nasr City for one summer; a blind check of the material map.

See more

Summer hyperspectral scenes (EnMAP, EMIT, Satellite 813); night-time heat from ECOSTRESS.

Map every building

Google Open Buildings and Microsoft footprints fill OpenStreetMap's gaps in Cairo.

Protect the vulnerable

Add age (over-65s, under-5s) and access to cooling; a heat vulnerability index.

Scale across MENA

Same free data and code for the next cities on our globe: Riyadh, Amman, Casablanca.

Add costs

Price per block for each measure from Egyptian supplier quotes, so planners compare °C per pound.

01 // VIEW

MATERIALS

SURFACE HEAT

HEAT RISK

GROWTH

Land surface temperature per 90 m block: median of summer 2023–25 Landsat 8/9 scenes. Surface, not air, temperature.

ANALYSIS MAPS

02 // MODEL NOTES

[+]

04 // CON

LEFT-DRAG
RIGHT-DRAG
CTRL+Z
ALT+DRAG
W A S D
F11

SURFACE TEMP · AVG

44.7 °C

today · mean of 576 blocks

HEAT EXPOSURE

6,460

person·°C · no plan yet

RESIDENTS COOLED

0

of 95,141 residents

[COOLCAIRO // TEAM 46 · EGYPT]

Thank you

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TEAM LEAD

Eng. Mohamed Esmat

TECHNICAL LEAD

Eng. Mahmoud Mostafa

DATA & HYPERSPECTRAL ANALYST

Code, notebooks, methodology report and the Windows app: github.com/mohamedEsmat/CoolCairo

Demo video: youtu.be/jkDC3HfMiQc

Data: Landsat (USGS) · Contains modified Copernicus Sentinel data [2023–2025] · Contains modified EnMAP data © DLR [2025] · WorldPop 2024 (CC BY 4.0) · © OpenStreetMap contributors · Google Open Buildings 2.5D Temporal (CC BY 4.0) · NASA Blue Marble

[ROOF]

COOL ROOFS

[TREE]

STREET TREES

[ROAD]

COOL PAVEMENT

[PARK]

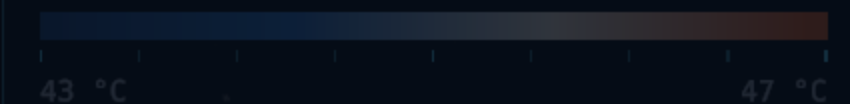
POCKET PARKS

UNDO

RESET ALL

Drag over blocks to paint, right-drag to erase. Coat every roof in a block white. Per unit of area coated: -4.6 °C on dark roofs, -3.6 °C on pale roofs (published values).

Land surface temperature (grey = typical block, 45.7 °C)



COLOURS

REPORT

INFERNO

WARM